Empathy in Real vs. Synthetic Clinician Comments — Cleaned/Balanced Data & Bayesian Models

0.1 Data cleaning

Two changes were made to the synthetic comments before any modeling, applied in this order (see `3e_filter_and_balance.py` for the full implementation):

1. **Length-outlier removal.** Several candidate definitions were compared on synthetic comment length before picking one (Table 1); **MAD-based modified z-score on log(length), threshold 3.5, computed separately within each (model, method) group** was chosen.
2. **Per-thread balancing.** For each (post_id, model, method) cell, synthetic comments are randomly subsampled down to exactly that thread's real comment count, so every group is compared against real using the same per-thread sample size rather than synthetic being systematically overrepresented. Real comments are never touched by either step.

The two interact: outlier removal happens first, so a cell can end up with *fewer* synthetic comments than its thread's real count. Per an explicit decision, such a cell simply keeps whatever remains rather than being dropped or backfilled - no thread is removed and no comment is invented.

Across the six candidates tested, the share of synthetic comments excluded ranges from 0.029% (iqr_3.0_log) to 4.46% (percentile_2.5_97.5_log). The chosen method (highlighted) excludes 332 comments (0.26%), causing 105 of 59384 (post_id, model, method) cells (0.18%) to fall short of their thread's real count under the "keep whatever remains" rule.

0.1.1 Is the exclusion rate the same across (model, method) groups?

No - a chi-square test of independence between exclusion (yes/no) and (model, method) group rejects equal proportions decisively ($\chi^2 = 184.8$, df = 8, p 1e-35). The exclusion rate ranges from 0.000% (Openai/DG) to 0.608% (Gemini/SAGII) of a group's comments - concentrated in specific (model, method) combinations rather than spread evenly.

0.1.2 Does the excluded/affected data carry empathy signal?

For the 82 (post_id, model, method) cells emptied to zero synthetic comments, both the excluded comments themselves and the real comments on the same threads were checked for whether they carried any empathy signal (label 0 = none vs. label 1 or 2 = some), per mechanism:

Both sides are predominantly label-0 across all three mechanisms - the share carrying any empathy signal (label > 0) ranges 3.0% to 14.9% depending on mechanism and side. Emptying these cells mostly discards

Table 1: Candidate length-outlier definitions tested on synthetic comments, before choosing one.

Method	Comments excluded	% excluded	Shortfall cells	% of cells
iqr_1.5_raw	1262	0.981	622	1.047
iqr_1.5_log	1393	1.083	453	0.763
iqr_3.0_log	37	0.029	16	0.027
mad_3.5_log	332	0.258	105	0.177
percentile_1_99_log	2283	1.775	975	1.642
percentile_2.5_97.5_log	5732	4.455	2467	4.154

Table 2: MAD 3.5x exclusion and downstream shortfall, by (model, method) group.

Model	Method	N synthetic	N excluded	% excluded	Cells emptied to 0	% cells short
Gemini	DG	14301	68	0.475	0	0.000
Gemini	SAG	14288	41	0.287	7	0.167
Gemini	SAGII	14300	87	0.608	26	0.500
Grok	DG	14258	23	0.161	2	0.030
Grok	SAG	14300	34	0.238	15	0.318
Grok	SAGII	14300	57	0.399	25	0.440
Openai	DG	14302	0	0.000	0	0.000
Openai	SAG	14302	10	0.070	3	0.076
Openai	SAGII	14302	12	0.084	4	0.061

Table 3: Empathy label proportions among the comments lost to a fully emptied cell.

Side	Mechanism	N label = 0	N label > 0	N total	% label > 0
Excluded synthetic comments	Emotional Reactions	79	4	83	4.8
Excluded synthetic comments	Interpretations	77	6	83	7.2
Excluded synthetic comments	Explorations	73	10	83	12.0
Real comments on affected threads	Emotional Reactions	59	8	67	11.9
Real comments on affected threads	Interpretations	65	2	67	3.0
Real comments on affected threads	Explorations	57	10	67	14.9

comments with no empathy signal to begin with, though not entirely - a minority did carry some.

0.1.3 Resulting dataset

The final dataset has 8625 real and 77462 synthetic comments (59279 of 59384 cells exactly matched to their thread's real count; the remaining 105 are the outlier-caused shortfall cells above).

0.2 Empathy label proportions in the final dataset

Category proportions only (no means, since the labels are discrete none/weak/strong categories, not a continuous scale) - the share of comments at each empathy label, by group, per mechanism, in the cleaned/balanced dataset that the Bayesian models below will be fit to.

Emotional Reactions (% of comments at each label, by group):

Table 4: Synthetic comment count remaining after each cleaning step.

Step	N	Pct of original
Original synthetic	128653	100.0
After outlier removal	128321	99.7
After per-thread balancing	77462	60.2

Group	Label 0 (%)	Label 1 (%)	Label 2 (%)
real	85.3	12.3	2.4
gemini_dg	93.9	5.5	0.5
openai_dg	98.0	1.7	0.4
grok_dg	99.4	0.6	0.0
gemini_sag	92.9	6.0	1.1
openai_sag	96.4	2.4	1.1
grok_sag	98.8	1.1	0.1
gemini_sagii	92.8	6.3	1.0
openai_sagii	96.3	2.6	1.1
grok_sagii	99.4	0.6	0.0

Interpretations (% of comments at each label, by group):

Group	Label 0 (%)	Label 2 (%)
real	89.3	10.7
gemini_dg	96.3	3.7
openai_dg	89.2	10.8
grok_dg	89.7	10.3
gemini_sag	97.0	3.0
openai_sag	86.2	13.8
grok_sag	89.2	10.8
gemini_sagii	96.0	4.0
openai_sagii	86.7	13.3
grok_sagii	87.5	12.5

Explorations (% of comments at each label, by group):

Group	Label 0 (%)	Label 2 (%)
real	81.1	18.9
gemini_dg	94.8	5.2
openai_dg	98.8	1.2
grok_dg	98.3	1.7
gemini_sag	94.4	5.6
openai_sag	99.4	0.6
grok_sag	97.5	2.5
gemini_sagii	95.2	4.8
openai_sagii	99.1	0.9
grok_sagii	98.7	1.3

0.3 Bayesian mixed-effects models: Methods

The modeling approach is unchanged from the earlier Bayesian companion report (`report_bayesian.Rmd`) - only the *data* changes, to the cleaned/balanced dataset described above. Recorded here now so the method is fixed before any results exist; the Results section below is filled in once the models are fit.

Motivation. On the uncleaned full dataset, Emotional Reactions' Model B (the model x method interaction) failed to converge under maximum likelihood - a non-positive-definite Hessian, consistent with quasi-separation in sparse cells. No mature Firth-style bias-reduction implementation exists for an ordinal cumulative-link outcome with a random intercept, in R or Python (checked directly). A Bayesian fit with weakly-informative priors resolves the same separation problem the same way Firth's correction does, while keeping the interaction and the `post_id` random intercept intact.

Models, identical formulas to the frequentist version, fit separately per mechanism:

- **A** (`label ~ group + log(length) + (1|post_id)`, all comments): each (model, method) group vs. real.

- **B** (`label ~ model * method + log(length) + (1|post_id)`, synthetic comments only): simple-effects contrasts between models within each method, and between methods within each model.

Families: Emotional Reactions uses `cumulative("logit")` (all 3 empathy levels); Interpretations and Explorations use `bernoulli("logit")` on `label == 2` (both never predict label 1 in this data).

Priors: `normal(0, 2.5)` on every fixed effect and (for Emotional Reactions) the ordinal thresholds - a standard weakly-informative default for logistic-type models (Gelman, Jakulin, Pittau & Su, 2008) - and `student_t(3, 0, 2.5)` on the `post_id` random-effect SD. brms's own default fixed-effect prior is an improper flat prior, which would not address the separation problem; the explicit prior above does.

Sampler: `brms`/Stan via `cmdstanr`, 4 chains, 2000 iterations (1000 warmup), within-chain threading enabled for speed. A fixed seed (20260101) is used throughout; because threading makes exact bit-for-bit reproducibility from a seed alone unreliable (verified directly: identical seeds with threading produce different draws, because multithreaded `reduce_sum` sums the log-likelihood in a non-deterministic order), reproducibility instead comes from caching every fitted model's complete posterior to disk (`brm(..., file = ...)`), not from being able to regenerate identical draws by rerunning the sampler.

Convergence is checked via max Rhat (< 1.01), minimum bulk-ESS ratio (> 0.1), and zero divergent transitions - reported per fit, not assumed.

Reporting: there is no p-value in a Bayesian fit. Contrasts are reported as odds ratios (link-scale estimates, exponentiated) with a 95% highest-posterior-density credible interval and `pd`, the posterior probability of direction. `pd > 0.975` is a rough two-sided-p$<$0.05 analogue (Makowski et al., 2019) but answers a different question, not a substitute; a contrast is only called credible when *both* `pd > 0.975` **and** the 95% CrI for the odds ratio excludes 1.

0.4 Results

Pending - the Bayesian models have not yet been fit on the cleaned/balanced dataset described above. This section will be filled in with the same convergence, vs.-real, and simple-effects tables as `report_bayesian.Rmd`, once `3c_bayesian_models.R` is rerun against `inference_results_filtered_balanced.csv`.